

Vector space: abelian gp, well-behaved mult by scalars in $\mathbb{R}$.

Words: linear combination, span, linear (in)dependence, basis, dimension

subspace Linear map $f: \text{vspace } V \longrightarrow \text{vspace } W$ means a set-function

Recall with $\left. \begin{aligned} f(v_1 + v_2) &= f(v_1) + f(v_2) \\ f(av_1) &= a f(v_1) \end{aligned} \right\}$ "preserves lines together with parametrization" (or collapses to a point)

Important observations Let $v_1, \dots, v_n$ be an ordered list of vectors in V , maybe with repeats

Define: $f: \mathbb{R}^n \longrightarrow V$ $\textcircled{2} f(e_i) = v_i$

$$\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \longmapsto \sum_{i=1}^n c_i v_i$$

✓ $\textcircled{1}$ f is a linear map

✓ $\textcircled{2}$ f is onto $\iff \text{span}(v_1, \dots, v_n) = V$ set w/o repeats,

✓ $\textcircled{3}$ f is 1-1 $\iff \{v_1, \dots, v_n\}$ is a linearly indep.

Upshot: The map $\mathbb{R}^n \xrightarrow{f} V$ is an isomorphism $\Leftrightarrow \{v_1, \dots, v_n\}$ is a basis B of V .
$$\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \mapsto \sum c_i v_i$$

In that case f^{-1} is called the coordinate map wrt the ordered basis B .

Notation If $f^{-1}(v) = [v]_B$ i.e. $[v]_B = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$ s.t. $v = \sum_{i=1}^n c_i v_i$.

Examples ① Is $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$ a basis for $\mathbb{R}^3$? $\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$ (check if cols are indep (equiv if they span $\mathbb{R}^3$)).
(NO) use table!

② $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ = basis for $\mathbb{R}^2$ $2c_2 - c_1 = c_3$

Find coord of following vectors wrt this basis: $\begin{bmatrix} 5 \\ 7 \end{bmatrix}, \begin{bmatrix} p \\ q \end{bmatrix}$

$$Ax = b \\ \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

If $f: V \rightarrow W$ is an isomorphism then all linear algebraic statements that are valid in V can be converted to valid statements in W by applying f . E.g. If $B \subset V$ is a basis for V then $f(B)$ is a basis for W .

Why: ① If f is an injective linear map $f(\text{indep set})$ is indep.

② If f is a surj $\implies f(\text{spanning set})$ spans W .

This gives invariance of dimension for any $f: \text{dim } V$:
Given bases $|B_1| = n$ $|B_2| = k$
 $\mathbb{R}^n \simeq V \simeq \mathbb{R}^k$

Observe: Inverse of an isomorphism is also an isomorphism

Composition of two linear maps is linear !!
bijejective maps is bijective $\implies \left\{ \begin{array}{l} V \xrightarrow{f} W \xrightarrow{g} U, f \& g \text{ isom} \\ g \circ f \text{ is } \Downarrow \text{an isomorphism} \end{array} \right.$

- Next:
- How to convert (coord wrt) one basis into another basis of V
 - How to find matrix of linear $V \xrightarrow{f} W$ given bases $\begin{cases} B \text{ of } V \\ C \text{ of } W \end{cases}$
 - How to translate between $\left\{ \begin{array}{l} \text{matrix of } f \text{ wrt } B, C \\ \text{---//--- wrt other bases } B', C' \end{array} \right.$
 - Discover matrix multiplication!
 - Prove rank-nullity in abstract setting